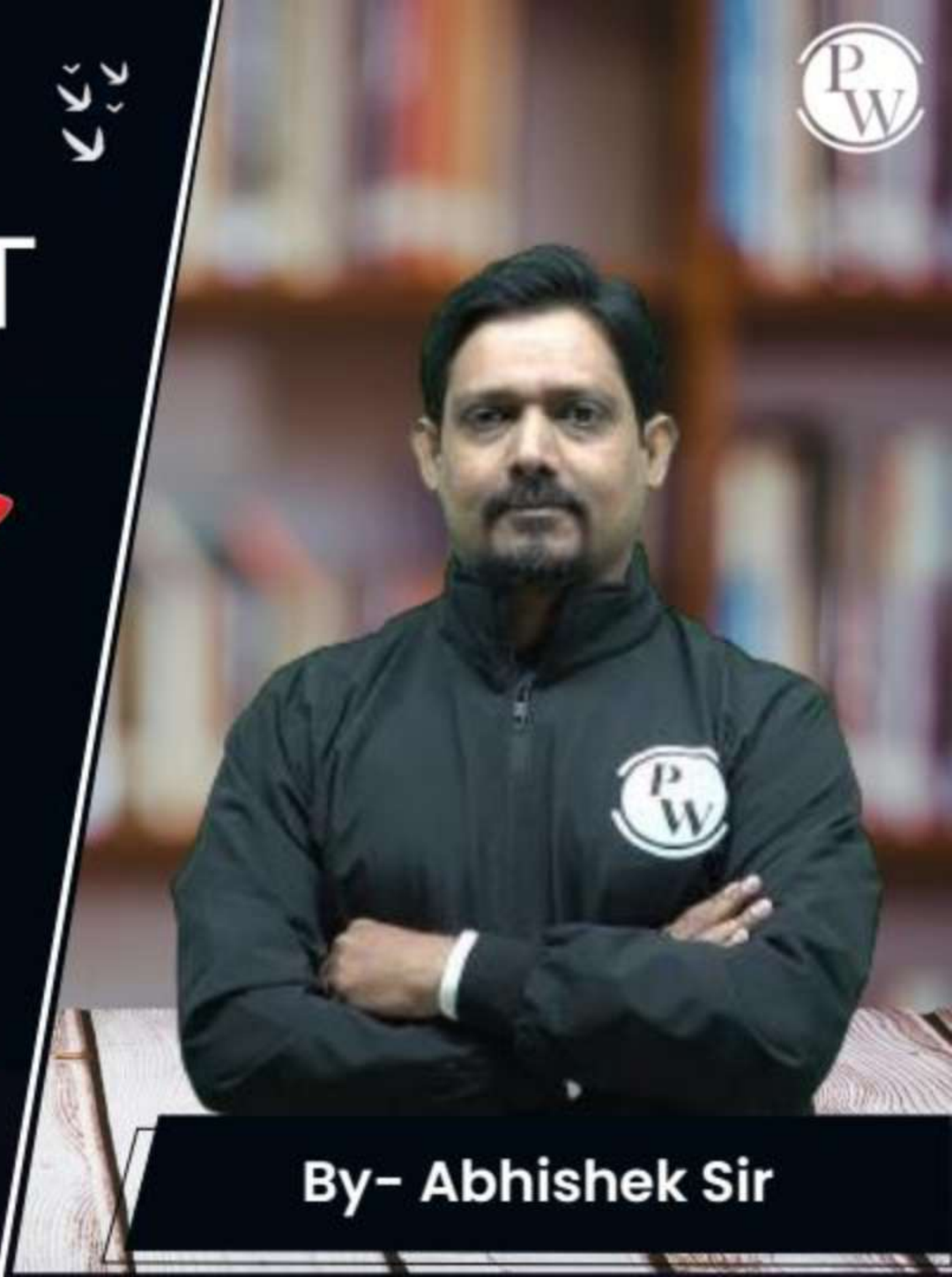
Computer Science & IT



C programming

Function & Storage Class

Lecture No. 01



By- Abhishek Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic
- Topic
- Topic

Continue

break

goto - Not advisable to use

Slide

Topics to be Covered



- Topic — function
- Topic — declaration, prototype
- Topic — Control transfer, control return
- Topic — Activation Record.
- Topic

Slide

Question

#Q.

```
#include<stdio.h>
int main()
{
   int i=0;
   for(i=2; i<=n; i*=2)
   printf("GATE WALLAH 2024");
   return 0;
}
```

The number of times printf() executed is approximated as-

(a) 2^n
(b) $\log_2 n$
(c) $\log_2 \log_2 n$
(d) 2

[B]

Slide



Question

```
#include <stdio.h>
int main() {
    int i,j=10, k=10;
        for(i=10;i<=200; i++){
            printf("\nLine 1");
        }
        do{
            printf("\nline 2");
            j = j+2;
        }while(j<=200);

       while(k<=200){
            printf("\nline 3");
            k = k+3;     }
         return 0 ;
```

200-10+1 = 191 X

$\frac{200-10+1}{2} = \lceil \frac{191}{2} \rceil = 96$

$\frac{191}{3} = 64$

The number of times line 1 printed is x , number of time line 2 printed is y and number of times line 3 printed is z then which of the following is TRUE ?

(A)x=y=z

(B)x>y>z ✓ x>y>z

(C)x<y<z

(D)x=y<z

Slide



Question

#Q.

```
#include <stdio.h>
int main() {
      int i=10;
      for (; i<=100; i++);{
            printf("\nI am a good Student");
      }
      printf("%d",i);
      return 0 ;
      }
```

Semicolon

101

101+1+1

=103

The value of I is x and number of print statement is y then value of x+y is

103

Slide



Question

#Q. The number of times loop executed is

```
#include<stdio.h>
int main(){
int i,count=0;
for (i=1; i<=100; i=i*2 ){
      printf("\nI am a good Student");

}
return 0 ;
}
```

The number of times value will be printed?

A. 6
B. 7
C. 8
D. 10

$\lfloor \log_2 100 \rfloor + 1$

$6 + 1 = 7$

7

Slide



Function

What is a function
Why function
Function execution
Function prototype
Declaration & definition
Function call
Parameter, argument
Return value



Function

```
int main() {
```

→ Name of function

→ return value.

{ } block of function

Every statement written in this block is part of main function and we can execute them by calling main fun^n

OS called the function

```
}
```

Slide



Function

1. logical group of instruction behaving as unit.

2. Big problem can be broken down to smaller units

   Reduces the complexity

3. Reusability : one time function created
   & we can reuse in different place.

Slide



Function

declaration : forward declaration

```
#include<stdio.h>
int fun(int,int);
```

⇐ prototype

⟶ int value return

⟶ Name of function fun

⟶ parameters

function will take 2 integer as argument



Function

```
#include <stdio.h>
int fun(int, int);
int main() {
   int a=10, b=20, c;
   c = fun(a,b);
   printf("%d", c);
   return 0;
}
```

Calling the function

function call

* When function is called control transfer to the called function.

Slide



Function

```
#include<stdio.h>
int fun(int,int);
int main(){
  int a=10, b=20, c;
  c=fun(a,b);
  printf("%d", c);
  return 0;
}
```

```
int fun(int x, int y){
  int z;
  z=x*y;
  return z; //200
}
```

x=a, y=b
x=10, y=20

// z=10*20
z=200

function definition

Actual parameter : a,b
formal parameter : x,y

Slide



Function

```
#include<stdio.h>
int fun(int,int);
int main(){
  int a=10, b=20, c;
  c = 200;
  printf("%d", c);
  return 0;
}
```

= print

```
int fun(int x, int y){
  int z;
  z = x*y;
  return z; //200
}
```

x=a, y=b
x=10, y=20

control transfer

control Return

z

// z=10*20
z=200

function definition

Actual parameter: a, b

formal parameter: x, y

Slide



Function

Control transfer :

1. Save the content in Activation Record then control transfer.
2. copy actual parameters to formal parameters
3. execute function body
4. once execution completed the control return back to the place from where function is called.
5. Resume execution of main (Here).

Slide



Function

6. The process of copying value of a and b to x and y : Call by value. (if parameters is present)

7. we can directly give definition of function before main()

Slide



Activation Tree and Records

Activation : A function is in execution

When a function executes the entire information about execution of function is maintained as record structure called Activation Record (file). When function start execution activation Record is created & Insert in Runtime stack.

Runtime stack

Slide



Activation Tree and Records

Local variable, return Address. content of Registers. ect.

Stack one ended data structure

Last in first out (LIFO)

Runtime stack

Slide

Activation Tree and Records

```
int main() {
    int a;
    f1();
    f2()
    :
}
```

```
① void f1() {
    int b;
    f2();
    :
}
```

```
void f2() {
    int c;
    :
}
```



Runtime stack

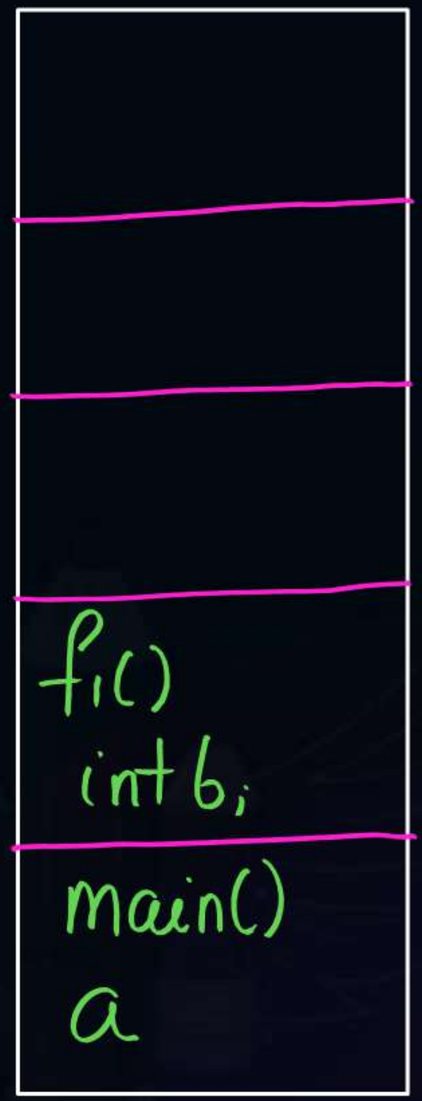
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Activation Tree and Records

```
int main() {
    int a;
    f1();
    f2()
    :
}
```

```
① void f1() {
    int b;
    f2();
    :
}
```

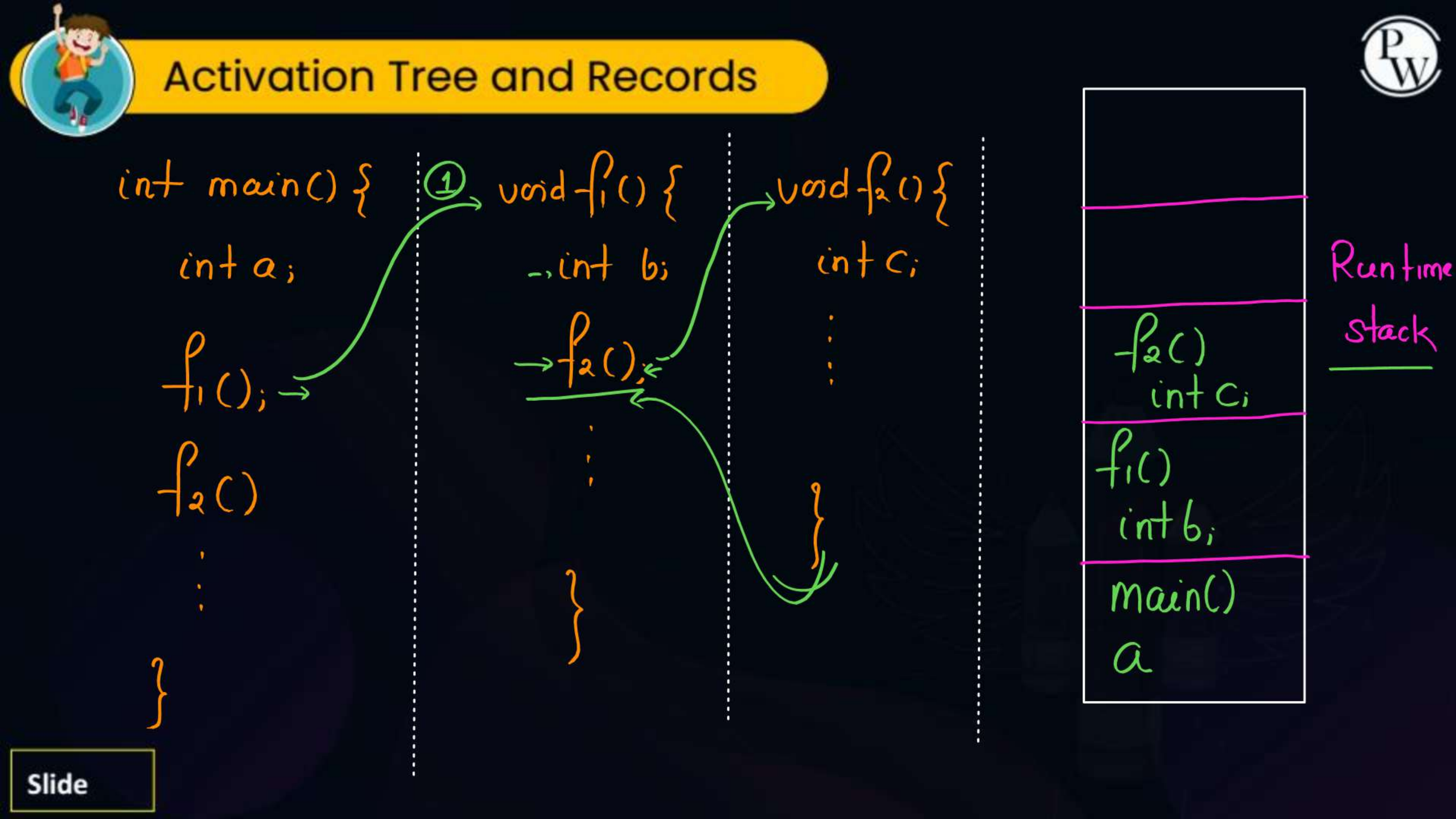
```
void f2() {
    int c;
    :
}
```



Runtime stack

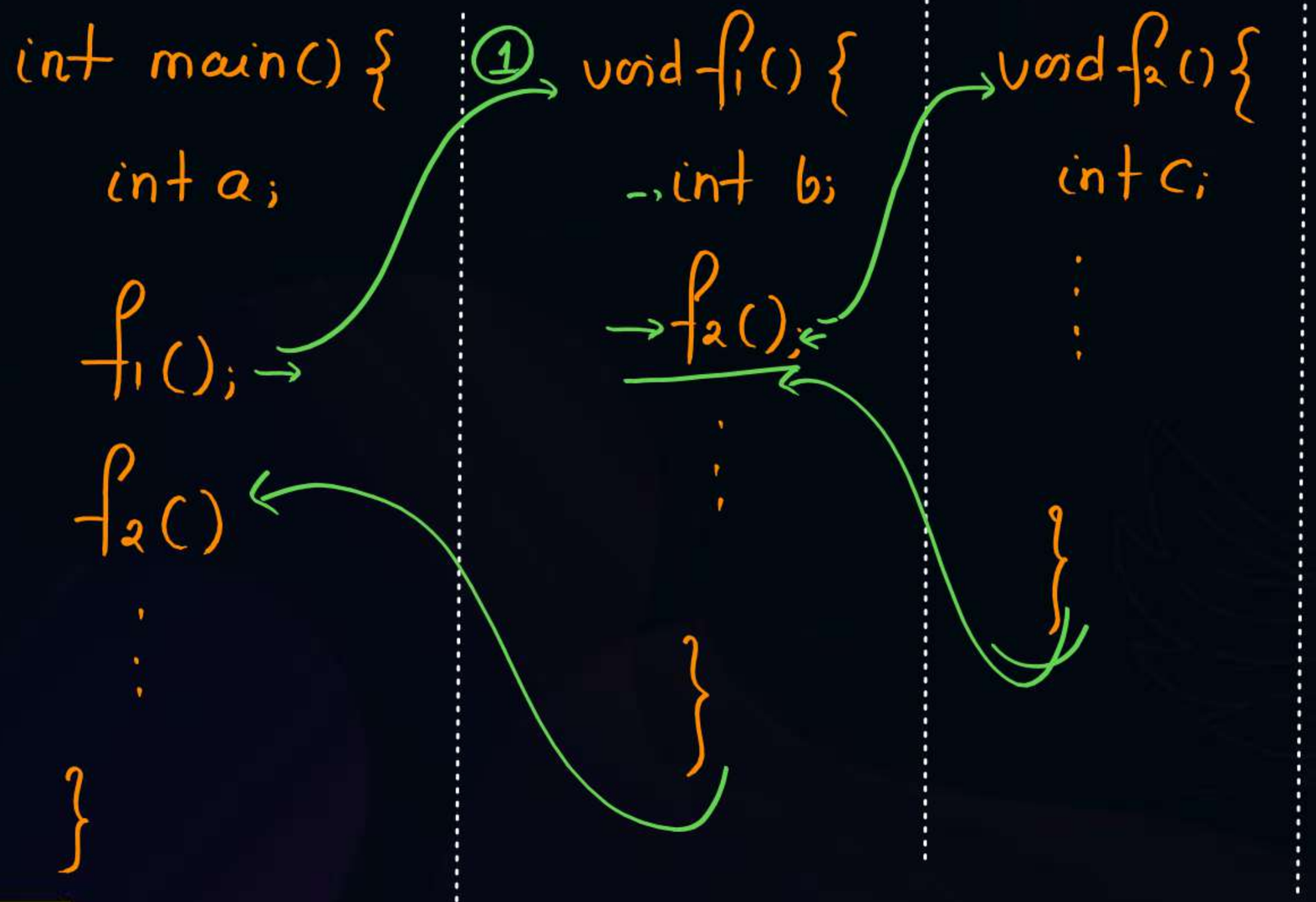
Slide

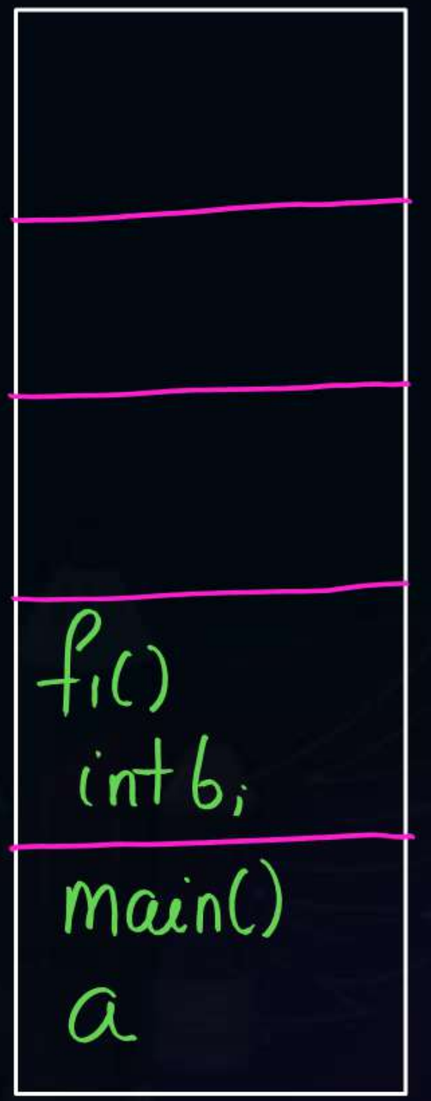
Activation Tree and Records



Slide

Activation Tree and Records

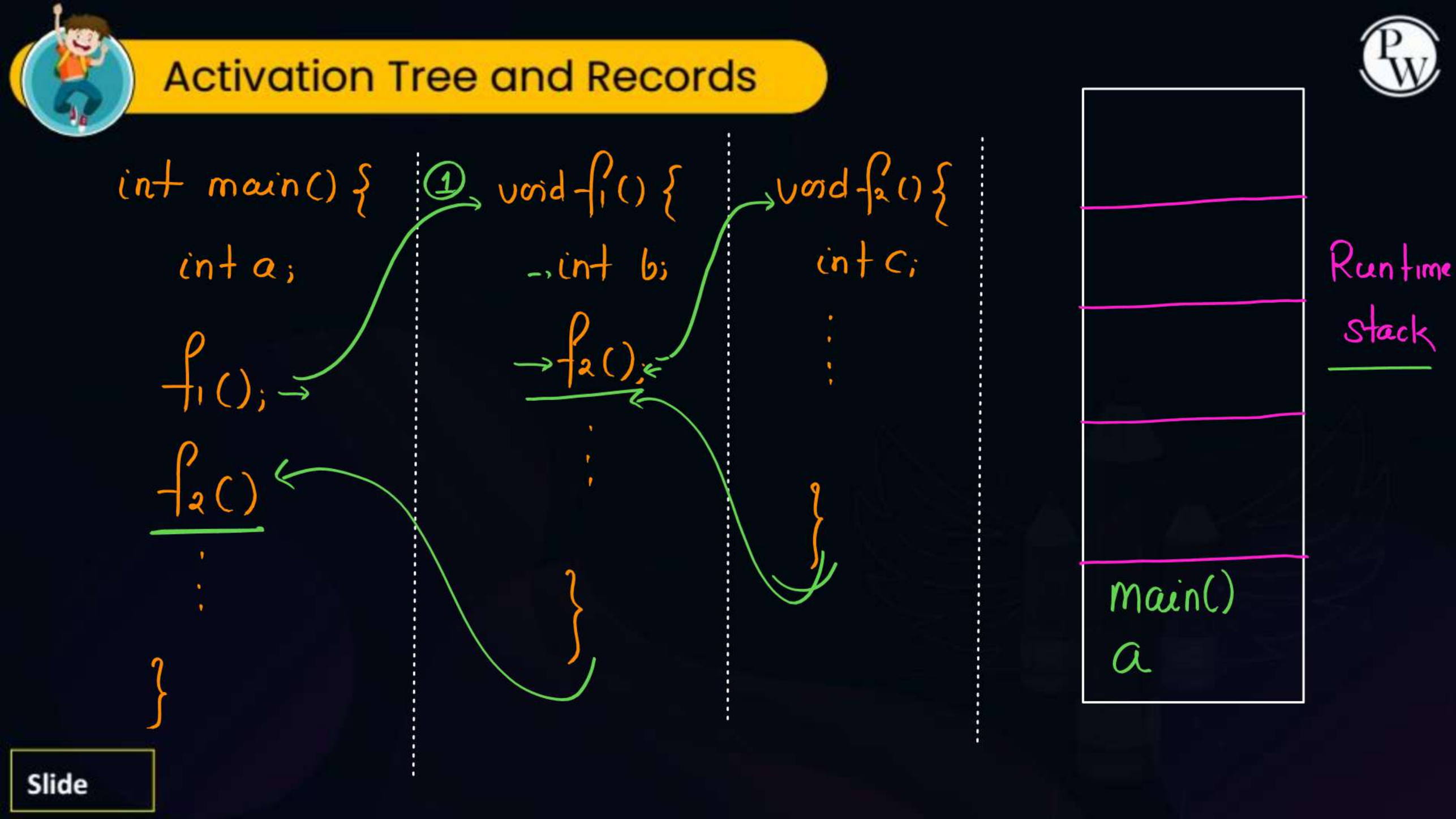




Runtime stack

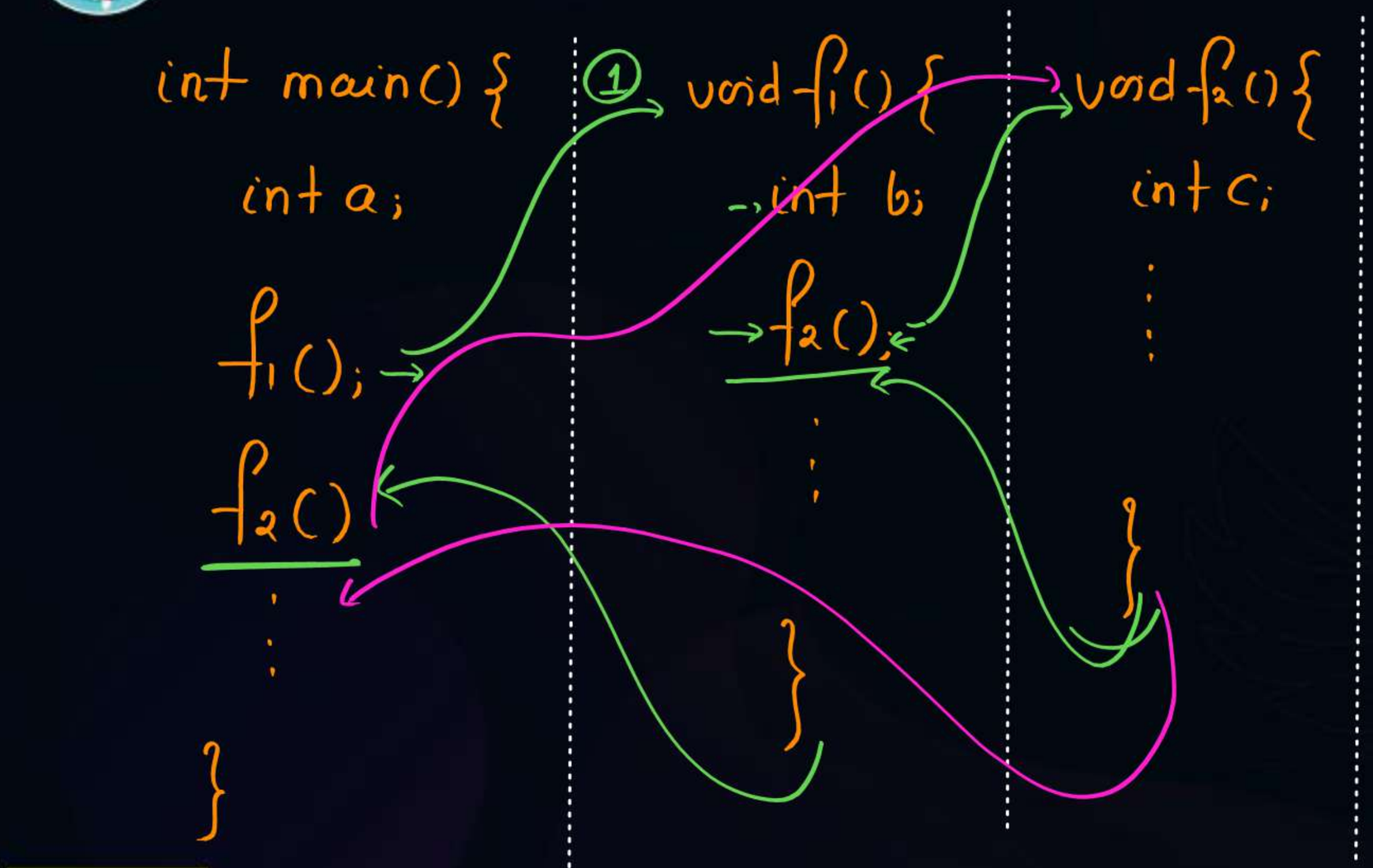
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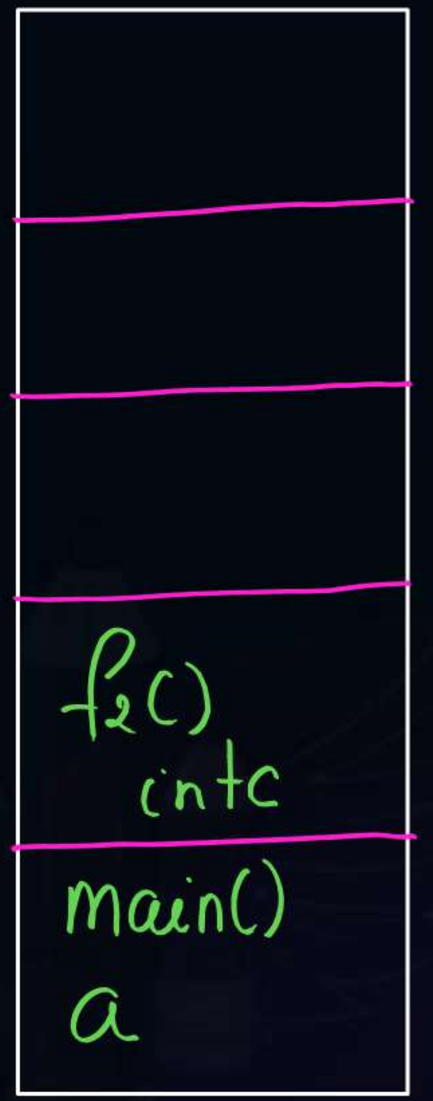
Activation Tree and Records



Slide

Activation Tree and Records

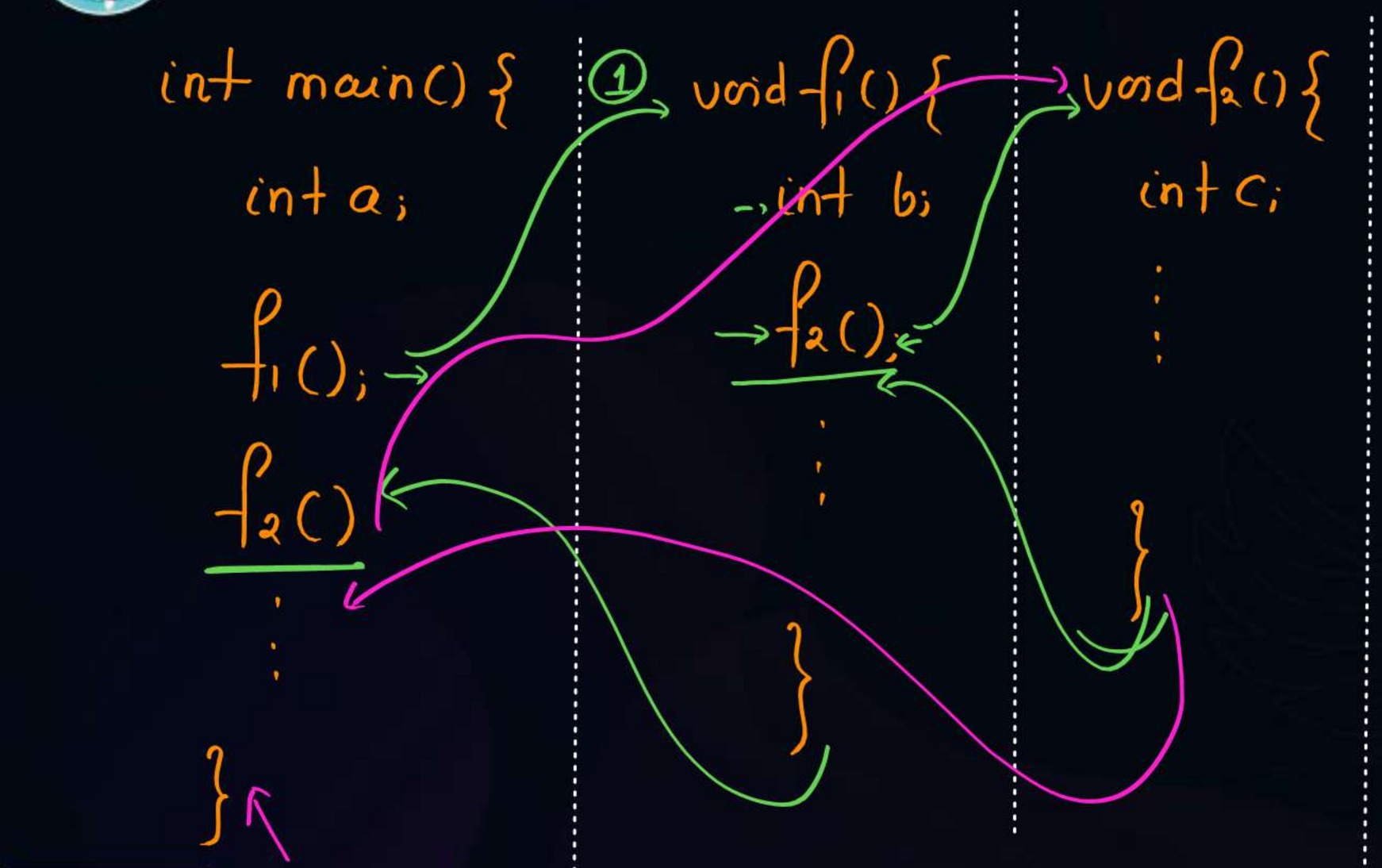




Runtime stack

Slide

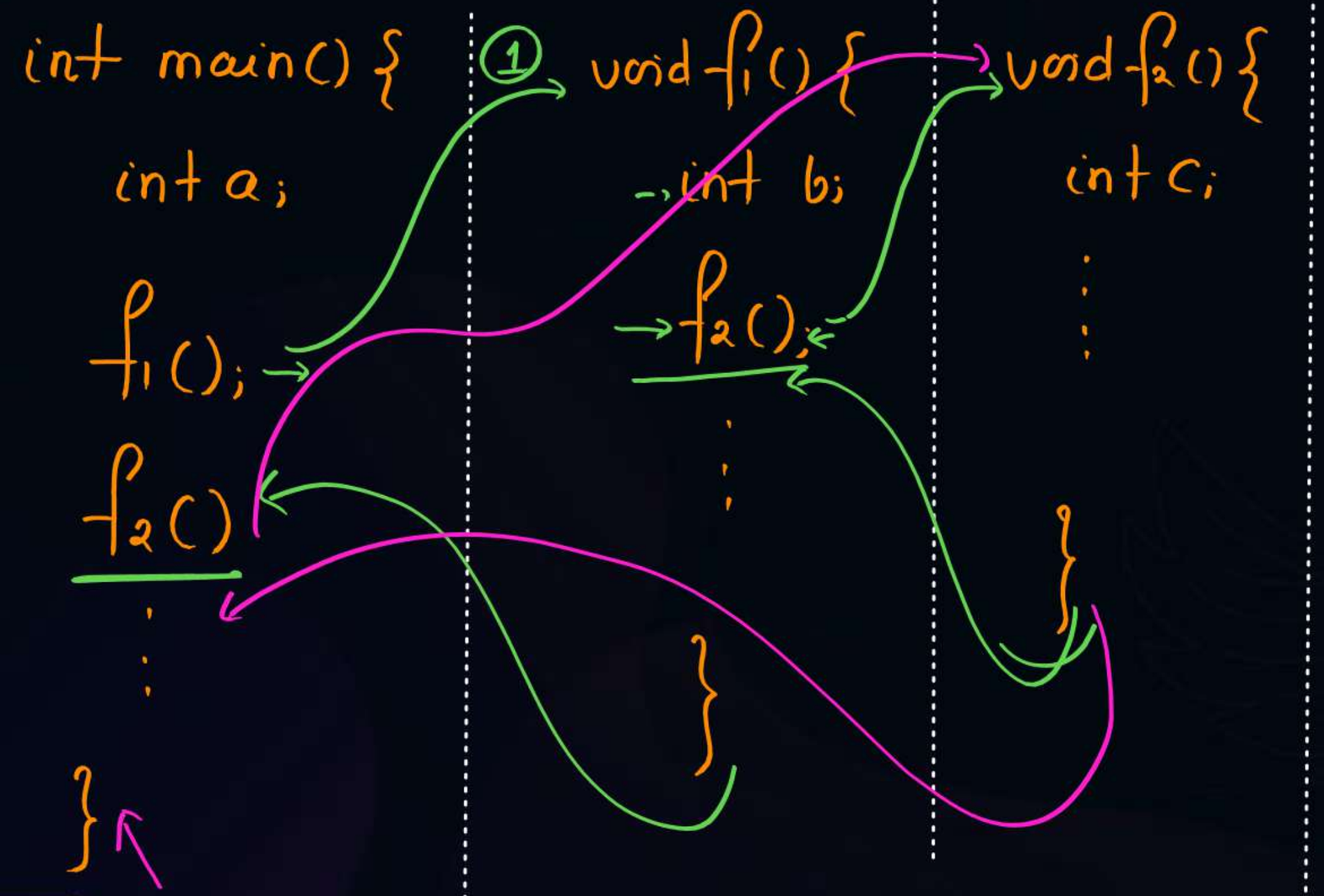
Activation Tree and Records





Slide

Activation Tree and Records





Runtime stack

Slide



Question

2.5 float forward declaration

#Q

No. of times '*' will be printed by the following C code is _____

```
#include<stdio.h>
void foo(int x)
{
        switch(x){
                case 1: printf("*");
                case 2: printf("*");
                case 3: printf("*");
                default: printf("*");
        }
}
int main()
{
        foo(2.5);
        return 0;
}
```

x = 2.5 → truncate → x = 2

definition before main

***

Number of star printed is ?

(A) 2

(B) 3 ✓

(C) 4

(D) 5

[B]

if definition is present then declaration is Not required

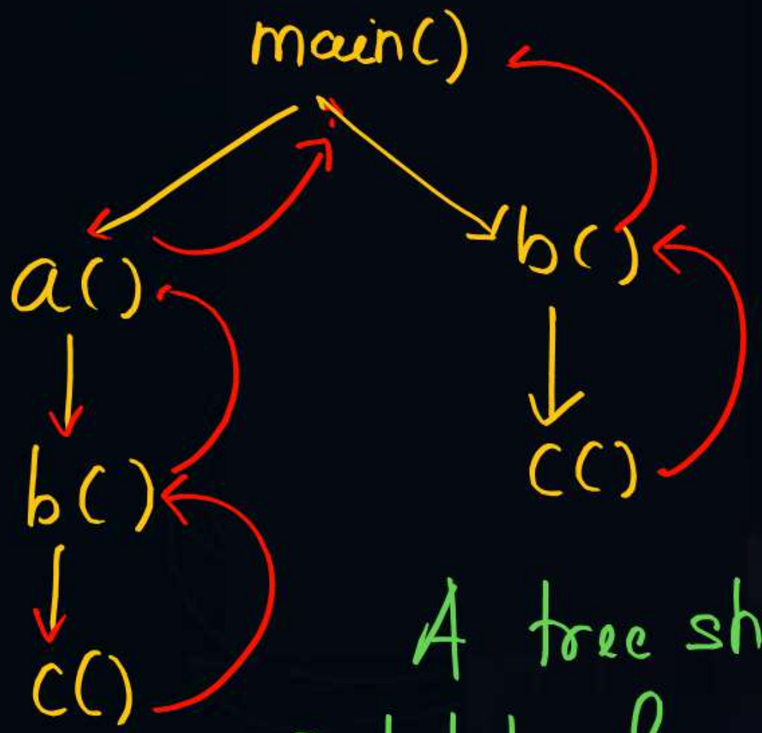
in place of variable yw directly pass value

Slide



Question

```
#Q The number of character printed by the code _____
#include<stdio.h>
void a();
void b();
void c();
int main(){
      a();
      b();
      return 0;
}
void a(){ printf("a"); b();}
void b(){ printf("a"); c();}
void c(){ printf("a");}
```



Activation tree

A tree showing control transfer and return of different activation Record is called Activation tree

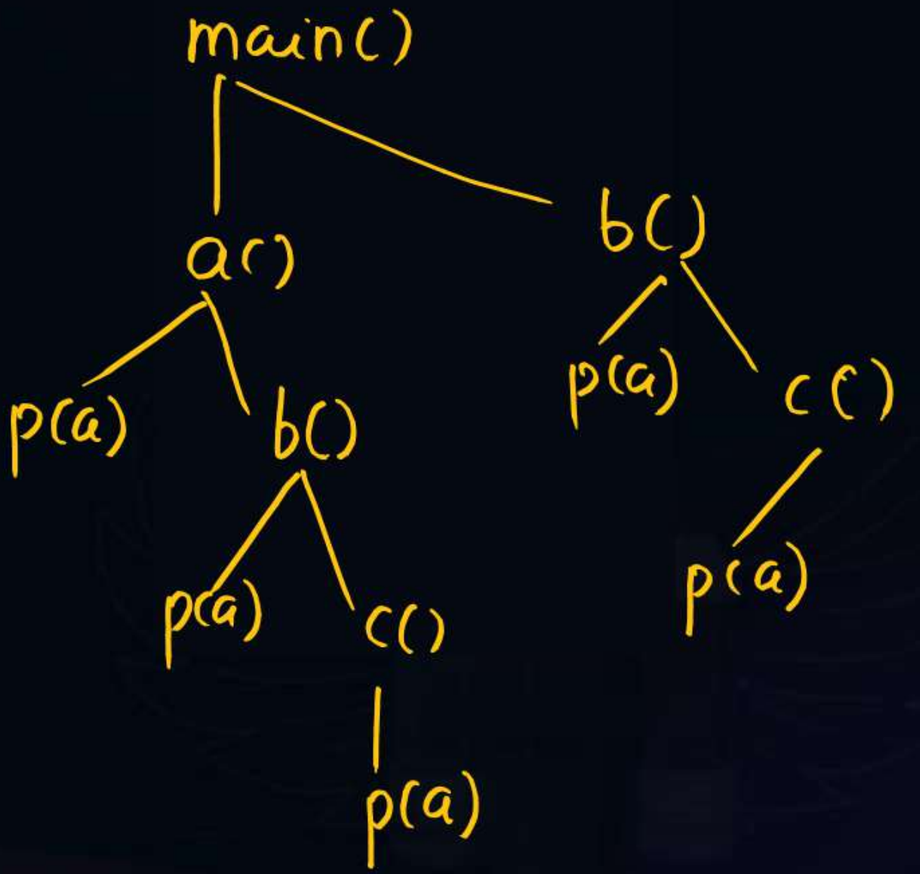
Slide



Question

#Q The number of character printed by the code _____

```
#include<stdio.h>
void a();
void b();
void c();
int main(){
      a();
      b();
      return 0;
}
void a(){ printf("a"); b();}
void b(){ printf("a"); c();}
void c(){ printf("a");}
```



Slide



Question

#Q The number of character printed by the code _____

```
#include<stdio.h>
void a();
void b();
void c();
int main(){
      a();
      b();
      return 0;
}
void a(){ printf("a"); b();}
void b(){ printf("a"); c();}
void c(){ printf("a");}
```

(A) 2

(B) 3

(C) 4

(D) 5

Slide



Question

#Q Consider the following program

```
#include <stdio.h>
void a(){printf("1");}
void b(){a();printf("2");}
void c(){a();b();printf("3");}
void d(){a();b();c();printf("4");}

int main(){
   d();
}
```

What is the output of the following program?

(A) 11211234

(B) 11211243

(C) 11212134

(D) 11211324

Slide



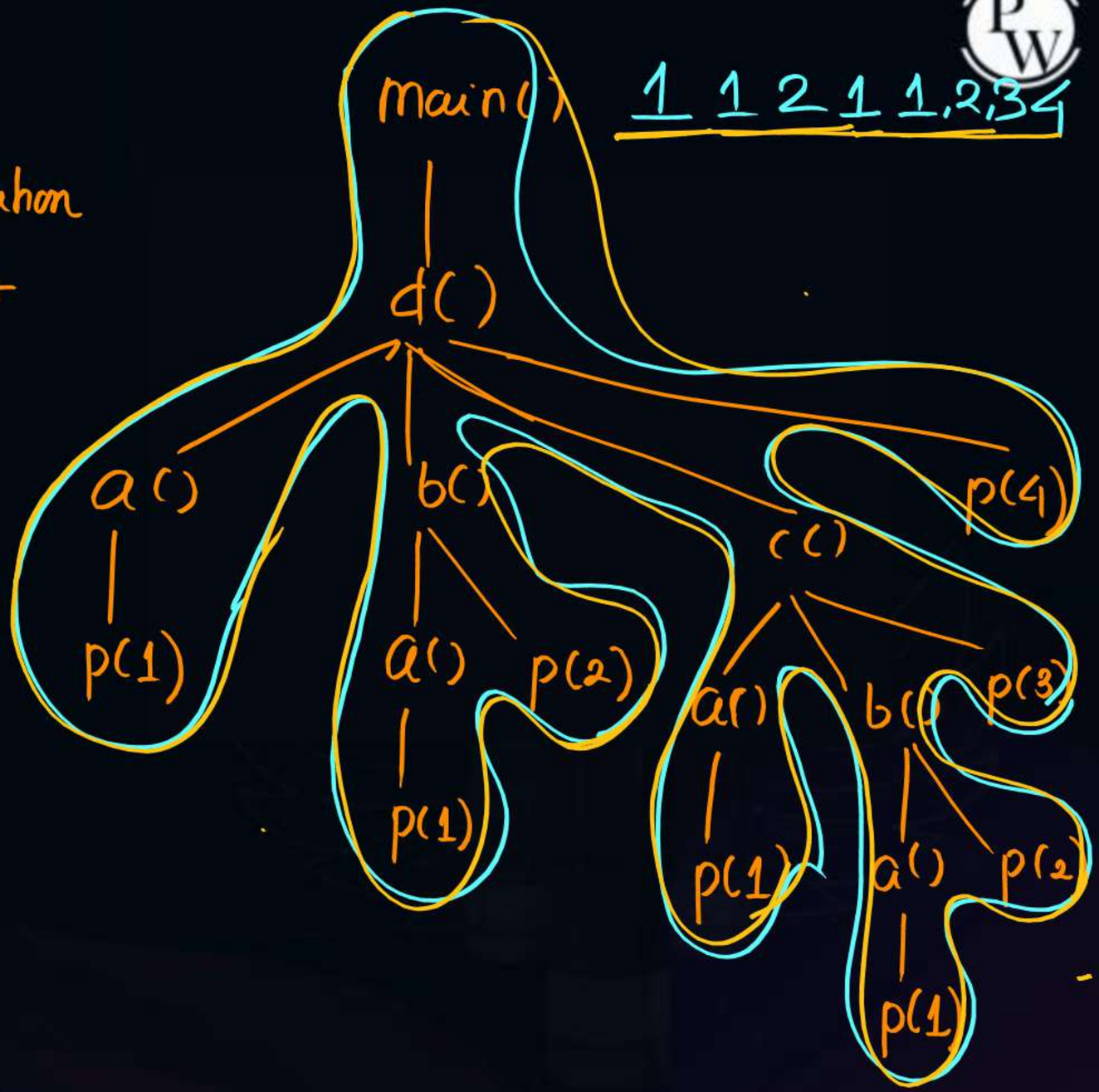
Question

flow of execution
left to right

#Q Consider the following program

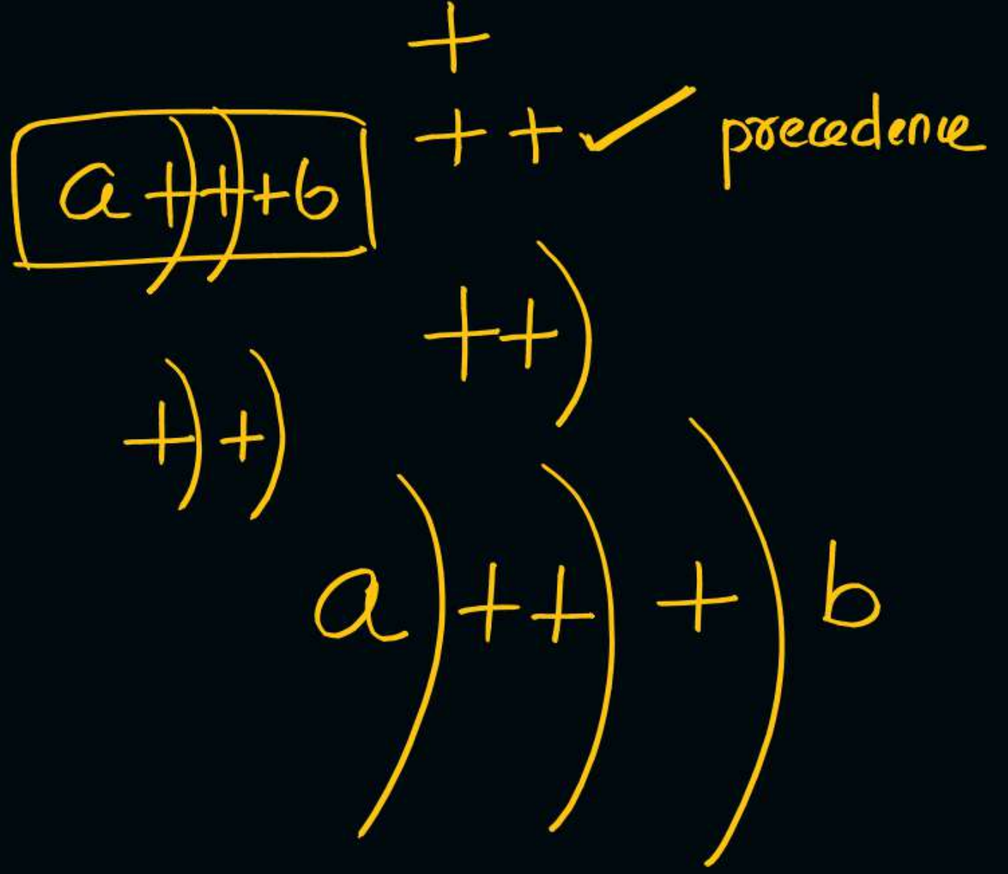
```
#include <stdio.h>
void a(){printf("1");}
void b(){a();printf("2");}
void c(){a();b();printf("3");}
void d(){a();b();c();printf("4");}

int main(){
   d();
}
```



1 1 2 1 1,2,3 4

Slide


a+++b
+
++ ✓ precedence
++)
+)+)
a)++)+)b



for loop

#Q.

```
#include <stdio.h>
int main()
{
int a,i;
a= 12.5/5+31.2/2*5-5;
for(i=a; i<90;i=i+3)
    {
      printf("%c\t", i+32);
    }
   return 0;
}
```

The output is-

(a) 107 110 113 116 119

(b) K L M N O

(c) k n q t w

(d) Garbage values

HW

Slide



Question

#Q.

Consider the following program:

```
#include <stdio.h>
int main()
{
    int i, j, n;
    for(i=1;i<=n;i++)
    for(j=1;j<=n;j++)
    printf("GATE Wallah");
    return 0;
}
```

How many times will the printf() statement be executed when n=32?

(a) 128 (b) 1024
(c) 512 (d) 256

Slide



2 mins Summary

Topic — function

Topic — declaration, control transfer

Topic — Activation Record & Activation

Topic — Tracing -

Topic

Slide

THANK - YOU

